Solve with us 6.1 - Not Graded

This assignment will not be graded and is only for practice.
Key Points: 1Key Points: 1

- A map $T: V \to W$ between two vector spaces V and W is said to be a linear transformation if for any two vectors v_1 and v_2 in the vector space V and for any scalar c in R the following conditions hold:

1. $T(v_1 + v_2) = T(v_1) + T(v_2)$
2. $T(cv_1) = cT(v_1)$

1 point
Let T be a function from R^2 to R^2 defined as follows:

$$T: R^2 \to R^2$$
$$T(x, y) = (2x, y)$$

Choose the set of correct option(s).
[Hint: Assume $v_1 = (x_1, y_1)$ and $v_2 = (x_2, y_2)$, then check the properties of linear transformation for the given function T.]

- [ ] $T(v_1 + v_2) = T(v_1) + T(v_2)$, for all $v_1, v_2 \in R^2$.
- [ ] $T(cv_1) = cT(v_1)$ for all $v_1 \in R^2$, and $c \in R$.
- [ ] T is a linear transformation.
- [ ] T is not a linear transformation.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$T(v_1 + v_2) = T(v_1) + T(v_2)$, for all $v_1, v_2 \in R^2$.
$T(cv_1) = cT(v_1)$ for all $v_1 \in R^2$, and $c \in R$.
T is a linear transformation.

1 point
Let T be a function from R^2 to R^2 defined as follows:

$$T: R^2 \to R^2$$
$$T(x, y) = (x - 1, y + 1)$$

Choose the set of correct option(s).
[Hint: Assume $v_1 = (x_1, y_1)$ and $v_2 = (x_2, y_2)$, then check the properties of linear transformation for the given function T.]

○ $T(v_1 + v_2) = T(v_1) + T(v_2)$, for all $v_1, v_2 \in R^2$.
○ $T(cv_1) = cT(v_1)$ for all $v_1 \in R^2$, and $c \in R$.
○ T is a linear transformation.
○ T is not a linear transformation.

No, the answer is incorrect.
Score: 0
Accepted Answers:
T is not a linear transformation.

Key Points: 2Key Points: 2

- A linear transformation $T: V \to W$ is said to be an isomorphism if it is bijective, i.e. the followings hold:

1. $T(v_1) = T(v_2)$ implies $v_1 = v_2$ (Injectivity).
2. For any $w \in W$ there exists $v \in V$ such that $T(v) = w$ (Surjectivity).

1 point
If $T: R^2 \to R^2$ is a linear transformation, such that $T(x, y) = (x, 0)$, then which of the following options is true?
[[Hints:

Hint 1:Hint 1: Let $v_1 = (x_1, y_1)$, $v_2 = (x_2, y_2)$, and assume that $T(v_1) = T(v_2)$. Then check whether $v_1 = v_2$ or not.
Hint 2:Hint 2: Does there exist any $v \in R^2$ for which $T(v) = (0, 1)$?]

- ◯ T is both one to one and onto.
- ◯ T is one to one but not onto.
- ◯ T is onto but not one to one.
- ◯ T is neither one to one nor onto.

No, the answer is incorrect.
Score: 0
Accepted Answers:
T is neither one to one nor onto.

1 point
If $T: R^2 \to R^3$ is a linear transformation defined by $T(x, y) = (2x + 3y, 5x - y, x + 6y)$. Which of the following options is true?
[Hint: Does there exist any $v \in R^2$ for which $T(v) = (1, 0, 0)$?]

- ◯ T is both one to one and onto.
- ◯ T is one to one but not onto.
- ◯ T is onto but not one to one.
- ◯ T is neither one to one nor onto.

No, the answer is incorrect.
Score: 0
Accepted Answers:
T is one to one but not onto.

Key Points: 3Key Points: 3

- Standard Ordered basis of R^n:

$$\{e_i \mid i = 1, 2, \ldots, n\},$$

where $e_1 = (1, 0, 0, \ldots, 0)$, $e_2 = (0, 1, 0, \ldots, 0)$, $\ldots$, $e_n = (0, 0, \ldots, 0, 1)$. In general, $e_i = (0, 0, \ldots, 0, 1, 0, \ldots, 0)$, where 1 is at the i-th position and 0 elsewhere.

- If $T(e_i)$ is known for all i, then the description of T can be known entirely. Suppose $T(e_i) = v_i$, for $i = 1, 2, \ldots, n$, then we have

$$T(x_1, x_2, \ldots, x_n) = T(x_1 e_1 + x_2 e_2 + \ldots + x_n e_n)$$

$$= T(x_1 e_1) + T(x_2 e_2) + \ldots + T(x_n e_n) = x_1 T(e_1) + x_2 T(e_2) + \ldots + x_n T(e_n) = x_1 v_1 + x_2 v_2 + \ldots + x_n v_n$$

1 point

Let $T: R^2 \to R^2$ be a linear transformation, such that $T(1, 0) = (3, 2)$ and $T(0, 1) = (1, 5)$. Which of the following options is true?

[Hint: $\{(1, 0), (0, 1)\}$ is the standard ordered basis of R^2.]

○ $T(x, y) = (3x + y, 2x + 5y)$

○ $T(x, y) = (3x + 2y, x + 5y)$

○ $T(x, y) = (5x, 6y)$

○ $T(x, y) = (4x, 7y)$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$T(x, y) = (3x + y, 2x + 5y)$

1 point

Let $T: R^2 \to R^2$ be a linear transformation, such that $T(2, 1) = (3, 2)$ and $T(1, 3) = (0, 0)$. Which of the following options is true?

[Hint: Express $(1, 0)$ and $(0, 1)$ in terms of $(2, 1)$ and $(1, 3)$.]

○ $T(x, y) = \frac{1}{5}(3x, 2y)$

○ $T(x, y) = (x + 1, y + 1)$

○ $T(x, y) = \frac{1}{5}(9x - 3y, 6x - 2y)$

○ T cannot be determined from the given information.

No, the answer is incorrect.
Score: 0
Feedback:

Accepted Answers:
$T(x, y) = \frac{1}{5}(9x - 3y, 6x - 2y)$

Key Points: 4

- Let $T : V \to W$ be a linear transformation. Let $\beta = \{v_1, v_2, ..., v_n\}$ be an ordered basis of V and $\gamma = \{w_1, w_2, ..., w_m\}$ be an ordered basis of W. So each $f(v_i)$ can be uniquely written as linear combination of w_j's, where $i = 1, 2, ..., n$ and $j = 1, 2, ..., m$.

$$T(v_1) = a_{11}w_1 + a_{21}w_2 + ... + a_{m1}w_m \quad T(v_2) = a_{12}w_1 + a_{22}w_2 + ... + a_{m2}w_m \quad \vdots \quad T(v_n) = a_{1n}w_1 + a_{2n}w_2 + ... + a_{mn}w_m$$

The matrix corresponding to the linear transformation T with respect to the ordered bases β and γ is given by,

$$\begin{bmatrix} a_{11} & a_{12} & ... & a_{1n} \\ a_{21} & a_{22} & ... & a_{2n} \\ \vdots & \vdots & & \\ a_{m1} & a_{m2} & ... & a_{mn} \end{bmatrix}$$

The above matrix not only depends on the linear map T, but also on the choice of the ordered bases β for V and γ for W.

- It is important to note that, for a fixed choice of basis of V and W, there is a one to one correspondence between the set of all linear transformations from V to W, and the set of all $m \times n$ matrices, where $n = \dim(V)$ and $m = \dim(W)$.

1 point
Suppose $T : R^2 \to R^2$ is a linear transformation, such that $T(x, y) = (x, 0)$. Which of the following matrices corresponds to T with respect to the standard ordered basis of R^2, i.e., $\{(1, 0), (0, 1)\}$, for both the domain and co-domain?
[Hint: $T(1, 0) = (1, 0)$ and $T(0, 1) = (0, 0)$]

○ $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

○ $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$

○

$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

○ $\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

1 point
Suppose $T: R^2 \to R^2$ is a linear transformation, such that $T(x, y) = (x + y, x - y)$. Which of the following matrices corresponds to T with respect to the standard ordered basis of R^2, i.e., $\{(1, 0), (0, 1)\}$, for the domain and $\{(1, 1), (1, -1)\}$ for the co-domain?
[Hint: $T(1, 0) = (1, 1)$ and $T(0, 1) = (1, -1)$]

○ $\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$

○ $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

○ $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

○ $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Feedback:
• T(1, 0) = (1, 1) = 1(1, 1) + 0(1, −1), and T(0, 1) = (1, −1) = 0(1, 1) + 1(1, −1).
Accepted Answers:
$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

1 point
Let $W = \{(x, y, z) \mid x = 2y + z\}$ be a subspace of R^3. Let $\beta = \{(2, 1, 0), (1, 0, 1)\}$ be a basis of W. Let $T: W \to R^2$ be a linear transformation such that $T(2, 1, 0) = (1, 0)$ and $T(1, 0, 1) = (0, 1)$. What will be the matrix representation of T with respect to the basis β for W and $\gamma = \{(1, 1), (1, -1)\}$ for R^2?

○ $\frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

○ $\frac{1}{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

○ $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

○ $\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

1 point
Let $\beta = \{(1,0),(0,1)\}$ and $\gamma = \{(1,1),(1,-1)\}$ be two bases of R^2. If $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is the matrix representation of a linear transformation $T : R^2 \to R^2$ with respect to β for the both the domain and co-domain. What will be the matrix representation of T with respect to γ for the domain and β for the co-domain?
[Hint: First find $T(x,y)$ and then find $T(1,1)$ and $T(1,-1)$.]

- $\begin{bmatrix} a+b & c+d \\ a-b & c-d \end{bmatrix}$
- $\begin{bmatrix} a-b \\ c-d \end{bmatrix}$
- $\begin{bmatrix} a+b & a-b \\ c+d & c-d \end{bmatrix}$
- $\begin{bmatrix} a+b & -a-b \\ c+d & -c-d \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Feedback:
• T(x, y) = (ax+by, cx+dy), so T(1, 1) = (a+b, c+d) and T(1, −1) = (a−b, c−d).
Accepted Answers:
$\begin{bmatrix} a+b & a-b \\ c+d & c-d \end{bmatrix}$